Q: Prove that function
$f(x) = 5x^4 + 7x + 3$ is $O(x^4)$

Answer:-

$$f(x) = 5x^4 + 7x + 3$$

Let, $g(x) = O(x^4)$

By the defination of Big-oh(O) we know that

$f(x) = O(g(x))$ iff

$f(x) \leq c \cdot g(x)$ for all $x \geq x_0$

where x_0 and c are constants where $c, x_0 > 0$

So, if we prove that

$$5x^4 + 7x + 3 \leq c \cdot x^4 \text{ for all } x \geq x_0$$

by finding c and n_0 values then we can say that:

$$5x^4 + 7x + 3 = O(x^4)$$

Let's find c:

$$5x^4 + 7x + 3 \leq c \cdot x^4$$

$$\Rightarrow 5 + \frac{7}{x^3} + \frac{3}{x^4} \leq c \cdot 1 \quad [\text{by multiplying } 1/x^4 \text{ both side}]$$

$$\Rightarrow 5 + 7 + 3 \leq c \quad [\text{by talking } x = 1]$$

$$\Rightarrow 15 \leq c$$

So, we can take $c = 15$ or grater than it any values after 15.

Now, let's put $c = 15$ and assume $x = 1$

$$\therefore \quad 5(1^4) + 7 \cdot (1) + 3 \leq 15 \cdot 1^4$$

$$\Rightarrow \quad 5 + 7 + 3 \leq 15$$

$$\Rightarrow \quad 15 \leq 15$$

the condition is satisfied by putting $c = 15$ and $x = 1$.

So, If we put $c = 15$ and $x \geq 1$ the definition will be true.

So, for all $x \geq 1$ and $c = 15$

$$5x^4 + 7x + 3 \leq c \cdot x^4$$

for,
$c = 15$
$x_0 = 1$
$x \geq 1$

Thus we can say that

$$\boxed{5x^4 + 7x + 3 = O(x^4)}$$

Q2:- Prove that function:

$$f(n) = 2n^3 + \frac{n^2}{4} + 100 \text{ is } \Theta(n^3)$$

Answer:

$$f(n) = 2n^3 + \frac{n^2}{4} + 100$$

Let, $g(n) = n^3$

By the defination of Theta (Θ) we know that

$f(n) = \Theta(g(n))$ iff

$f(n) \leq c_1 . g(n)$ and

$f(n) \geq c_2 . g(n)$ for all $n \geq n_0$

where n_0, c_1, c_2 are constants

where $n_0, c_1, c_2 > 0$

So, if we prove that

$$2n^3 + \frac{n^2}{4} + 100 \leq c_1 . n^3 \quad ---(1)$$

and

$$2n^3 + \frac{n^2}{4} + 100 \geq c_2 . n^3 \quad ----(2)$$

for all $n \geq n_0$, by finding c_1, c_2 and n_0 values then we can say that

$$2n^3 + \frac{n^2}{4} + 100 = \Theta(n^3)$$

Let's find c_1 from first condition:-

$$2n^3 + \frac{n^2}{4} + 100 \leq c \cdot n^3$$

$$\Rightarrow 2 + \frac{1}{4n} + \frac{100}{n^3} \leq c_1 \cdot 1 \quad [\text{by multiplying } 1/n^3 \text{ both sides}]$$

$$\Rightarrow 2 + \frac{1}{4} + 100 \leq c_1 \quad [\text{by taking } n=1]$$

$$\Rightarrow 102.25 \leq c_1$$

So, we can take $c_1 = 102.25$ or greater than any value of c for 102.25

Now, Let's put $c_1 = 102.25$ and assume $n = 1$

$$\therefore 2(1^3) + \frac{1^2}{4} + 100 \leq 102.25 \cdot 1^3$$

$$\Rightarrow 2 + \frac{1}{4} + 100 \leq 102.25$$

$$\Rightarrow 102.25 \leq 102.25$$

So the condition is satisfied by putting $c_1 = 102.25$ and $n = 1$

So, if we put $c_1 = 102.25$ and $n \geq 1$ the defination will be true

So, for all $n \geq 1$ and $c_1 = 102.25$

$$2n^3 + \frac{n^2}{4} + 100 \leq c \cdot n^3$$

1st condition proved

Let's find c_2 from 2nd condition:-

$$2n^3 + \frac{n^2}{4} + 100 \geq c_2 \cdot n^3$$

by calculation same way we can say

$$\Rightarrow \quad 102.25 \geq c_2$$

So, c_2 must be smaller than 102.25. (or equal)

Let' take smallest positive integer for c_2. $\boxed{c_2 = 1}$

Now, for $c_2 = 1$ and $n = 1$ we can check the 2nd condition is satisfied or not.

$$2 \cdot 1^3 + \frac{1^2}{4} + 100 \geq 1 \cdot 1^3$$

$$\Rightarrow \quad 2 + \frac{1}{4} + 100 \geq 1$$

$$\Rightarrow \quad 102.25 \geq 1$$

$\boxed{\begin{matrix} c_2 = 1 \\ n = 1 \end{matrix}}$

So, for $c_2 = 1$ and for all $n \geq 1$ the condition is satisfied

So, we can say that

$$2n^3 + \frac{n^2}{4} + 100 \geq c_2 n^3$$

[for $c_2 = 1$ and $n_0 = 1$ $n \geq 1$]

$\boxed{\text{2nd condition proved}}$

So, we ~~found~~ have found c_1, c_2 and n_0 to satisfy bothe the conditon of Theta (θ)

$$\begin{bmatrix} c_1 = 102.25 \\ c_2 = 1 \\ n_0 = 1 \end{bmatrix}$$

So, we can say that

$$2x^3 + \frac{x^2}{4} + 100 = \theta(x^3)$$